Department of

Information and Communication Technology

M.Sc. (Engg.) in ICT (2024-2025)

ICT 6127: Advanced TCP/IP Inter-networking

A Brief Intro on

Transmission Control Protocol (TCP)

Presented to **DR NAZRUL ISLAM**

Associate Professor, Department of ICT, MBSTU

Presented by

Rubaya Tamanna, **IT25642**

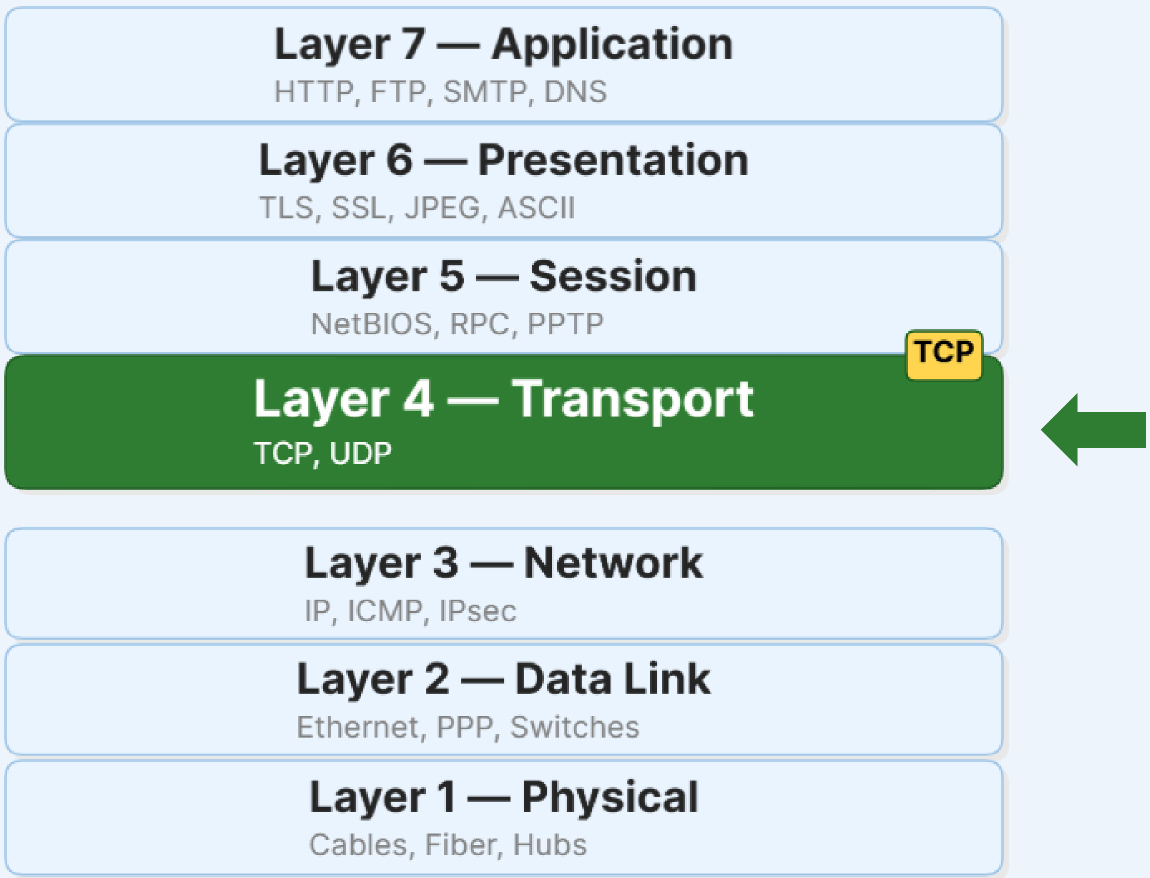
r.mistce15@gmail.com

GitHub
rubayamist.github.io

OUTLINE

- ❑ TCP in OSI and Its use in online payment
- ❑ Multiplexing And Demultiplexing
- ❑ Addressing and Header Format
- ❑ Connection And Termination
- ❑ State Transition
- ❑ Application And Limitations
- ❑ References

TCP IN OSI



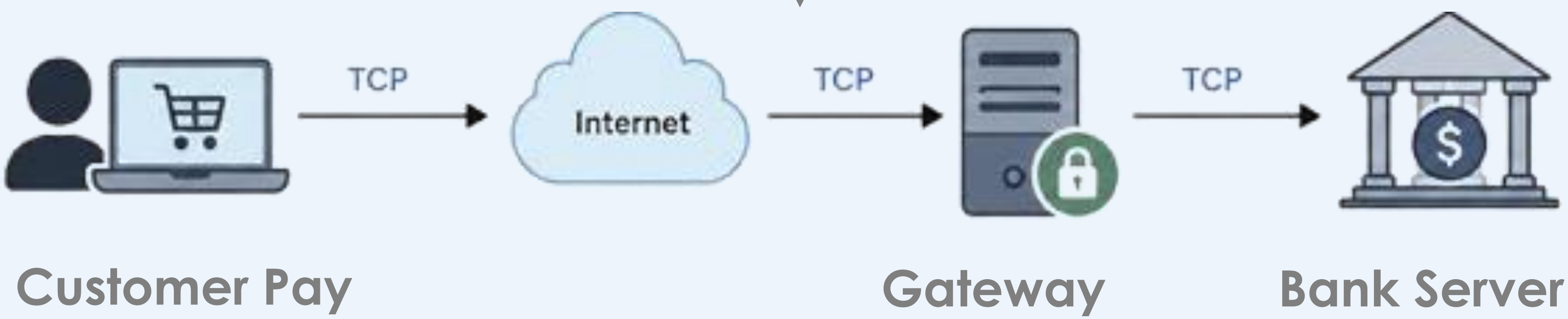
TCP: Transmission Control Protocol

- is a **connection-oriented transport layer protocol** that ensures reliable, ordered, and error-checked delivery of data between devices over a network.

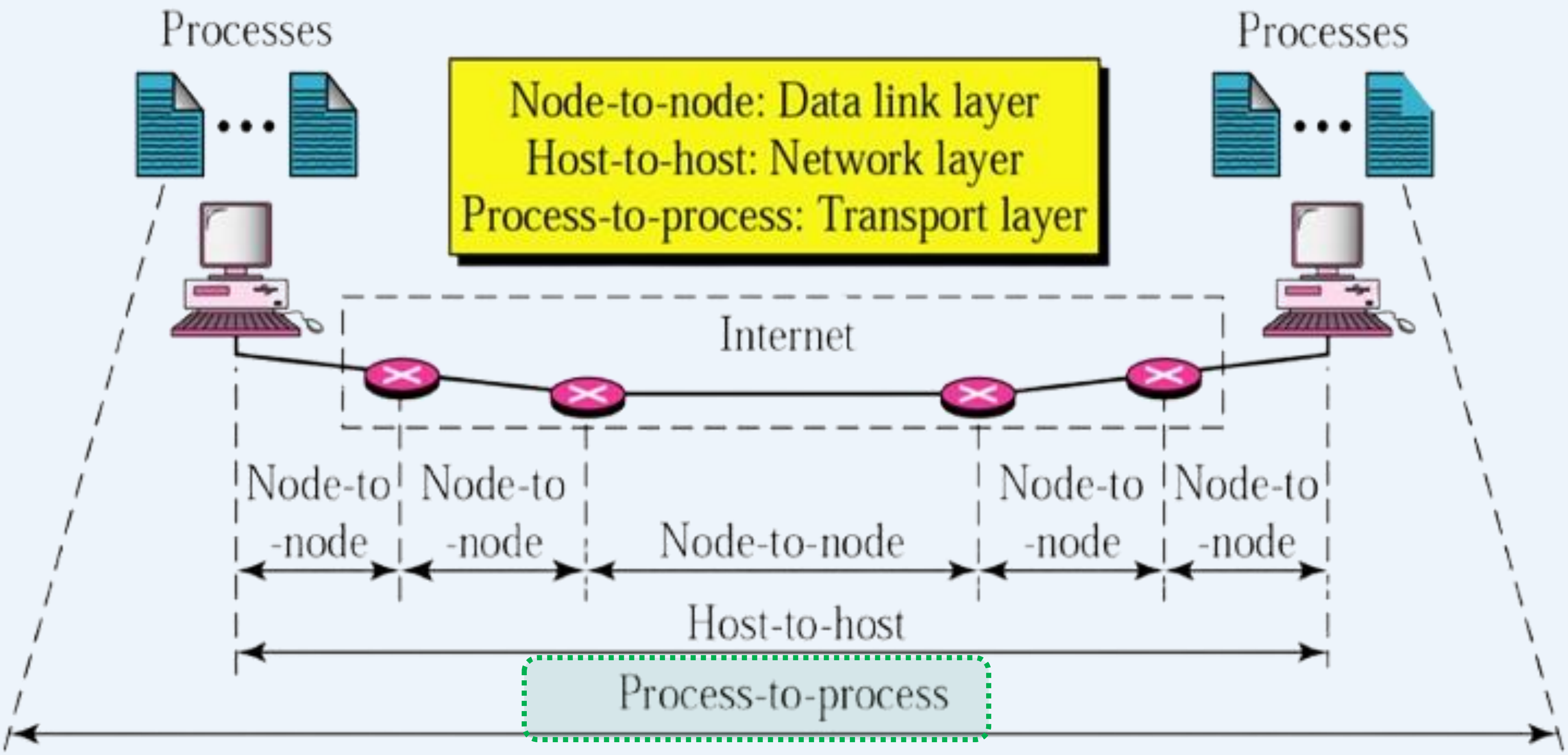
TCP IN BANKING

TCP Ensures:

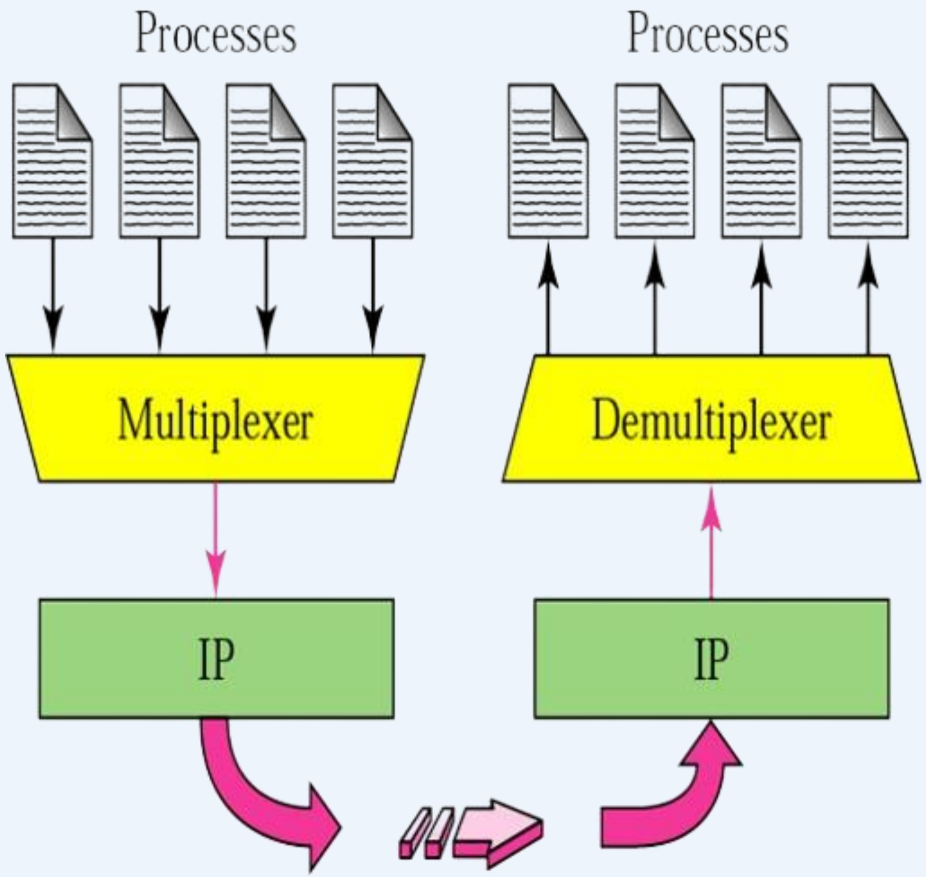
- Secure Connection
- and Reliable delivery
- Ordered data
- Error check and resend



TRANSPORT LAYER



MULTIPLEXING AND DEMULTIPLEXING



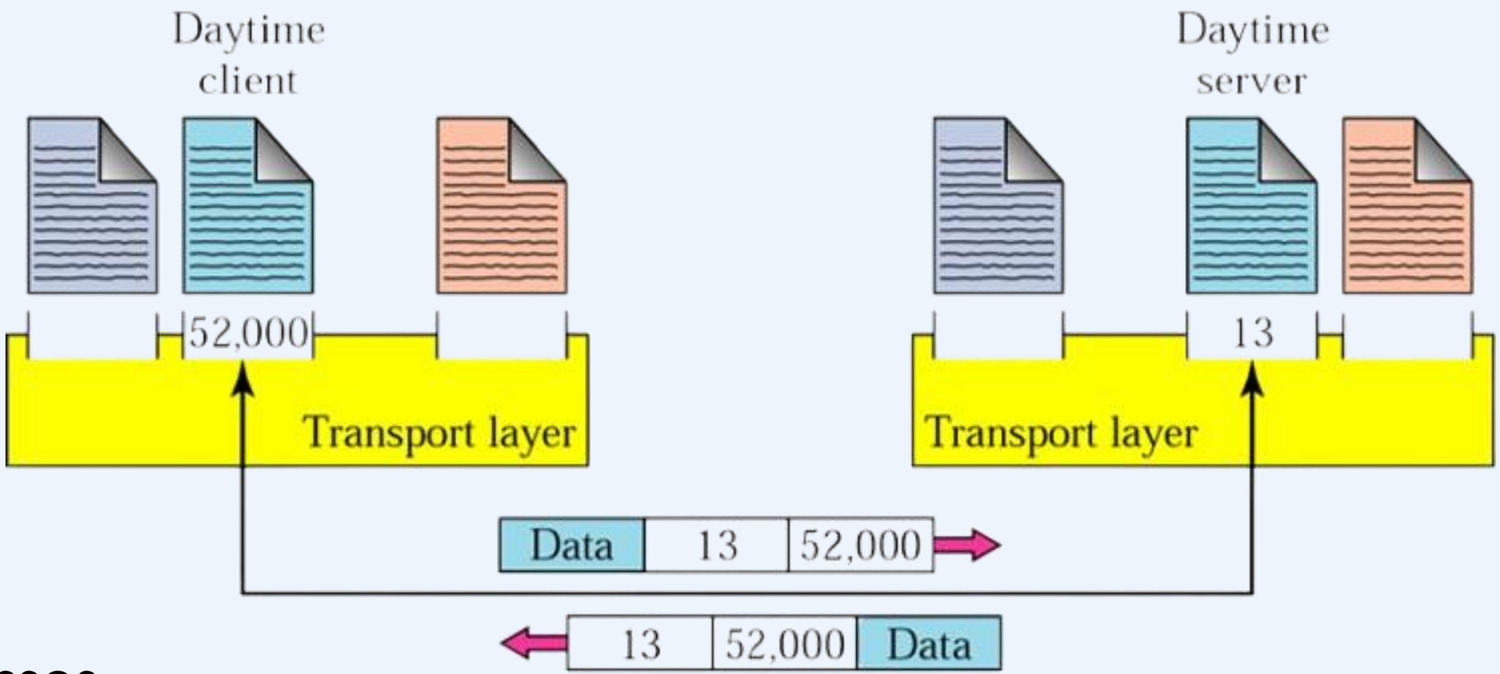
Multiplexing

- process of gathering data chunks from different processes at the sender
- ***Processes → TCP Protocol → Stream***

Demultiplexing

- Error checking
- Header removal
- Delivery
- ***Stream → TCP Headers → Processes***

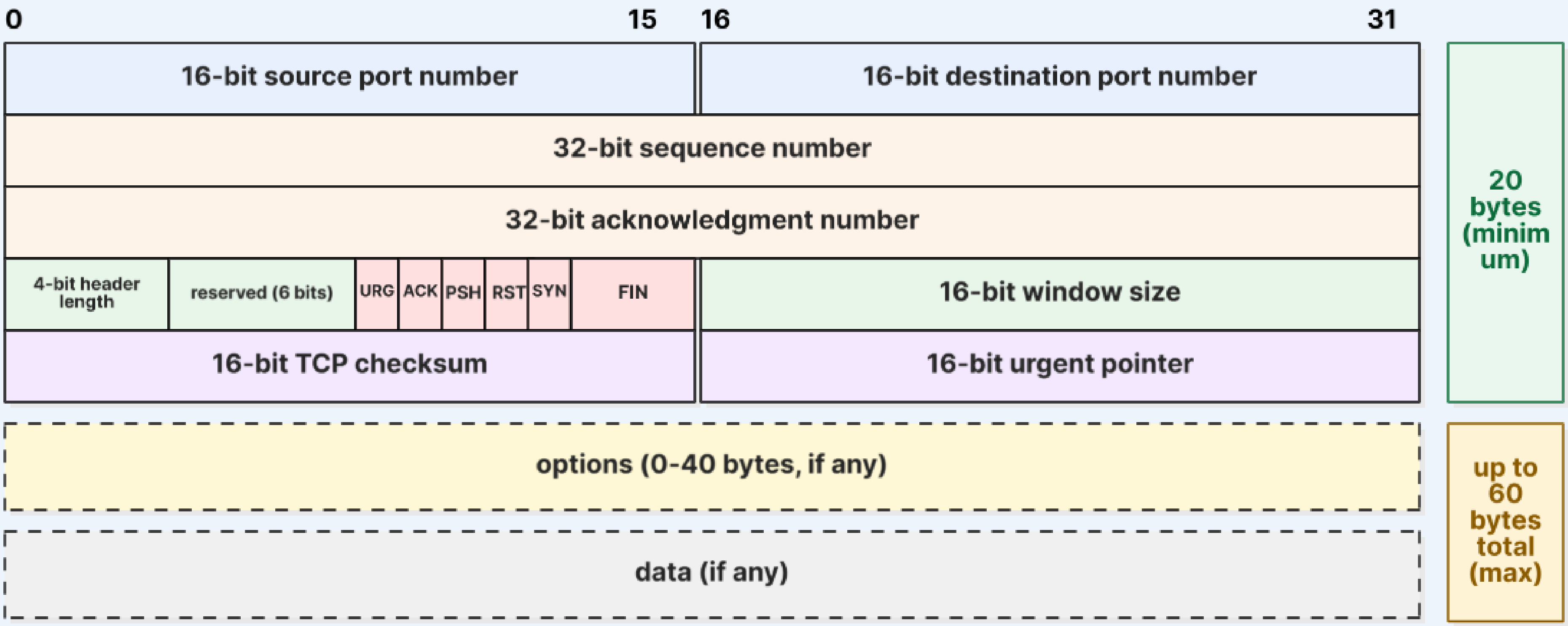
TCP ADDRESSING



Addresses

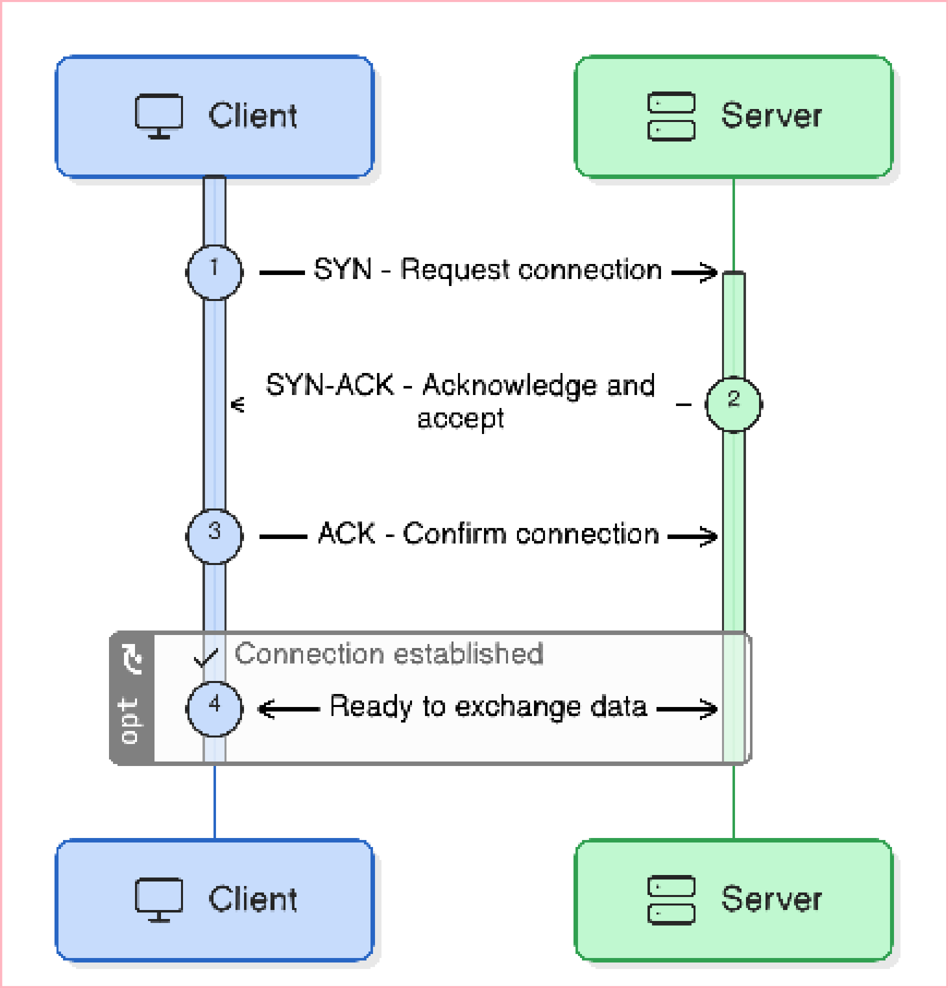
- Data link layer → MAC address
- Network layer → IP address
- **Transport layer → *Port number* (choose among multiple processes running on destination host)**

TCP HEADER FORMAT



- **Header length:** minimum **20 bytes** (no options) — **maximum 60 bytes** (*with up to 40 bytes of options*).
- The 4-bit *header length* field specifies the size in **32-bit words** (5–15).

TCP: CONNECTION

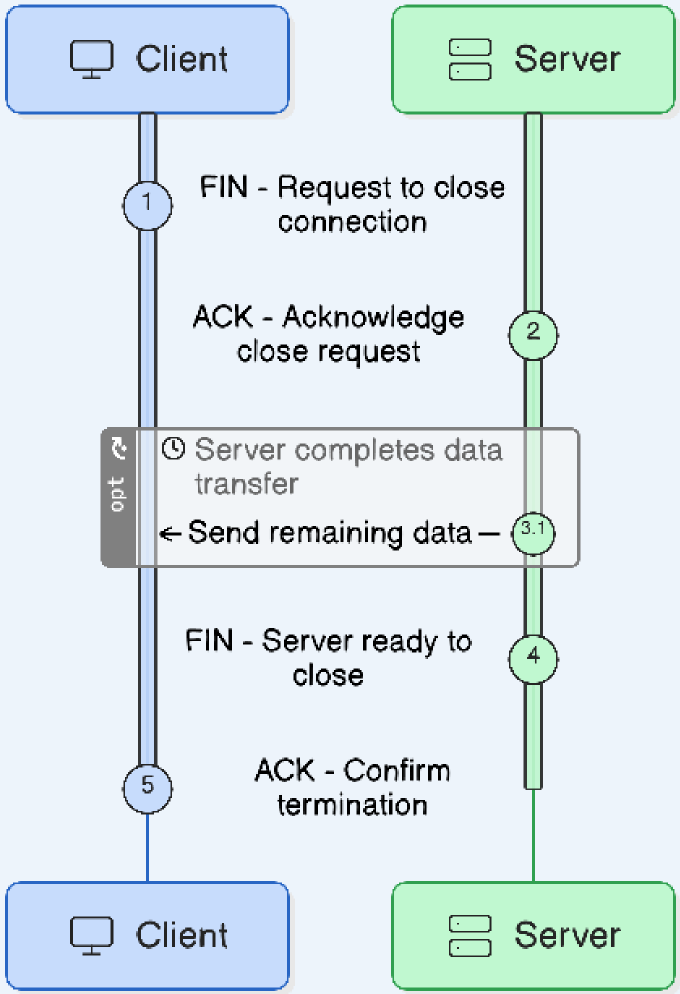


I. **SYN (Synchronize):** sender sends a SYN segment to connect

II. **SYN-ACK (Synchronize-Acknowledge):** receiver agrees to connect

III. **ACK (Acknowledge):** sender confirms connection established

TCP: TERMINATION



I. **FIN (Finish):** The sender who wants to close the connection sends a FIN segment to the receiver.

II. **ACK (Acknowledge):** The receiver acknowledges the FIN with an ACK.

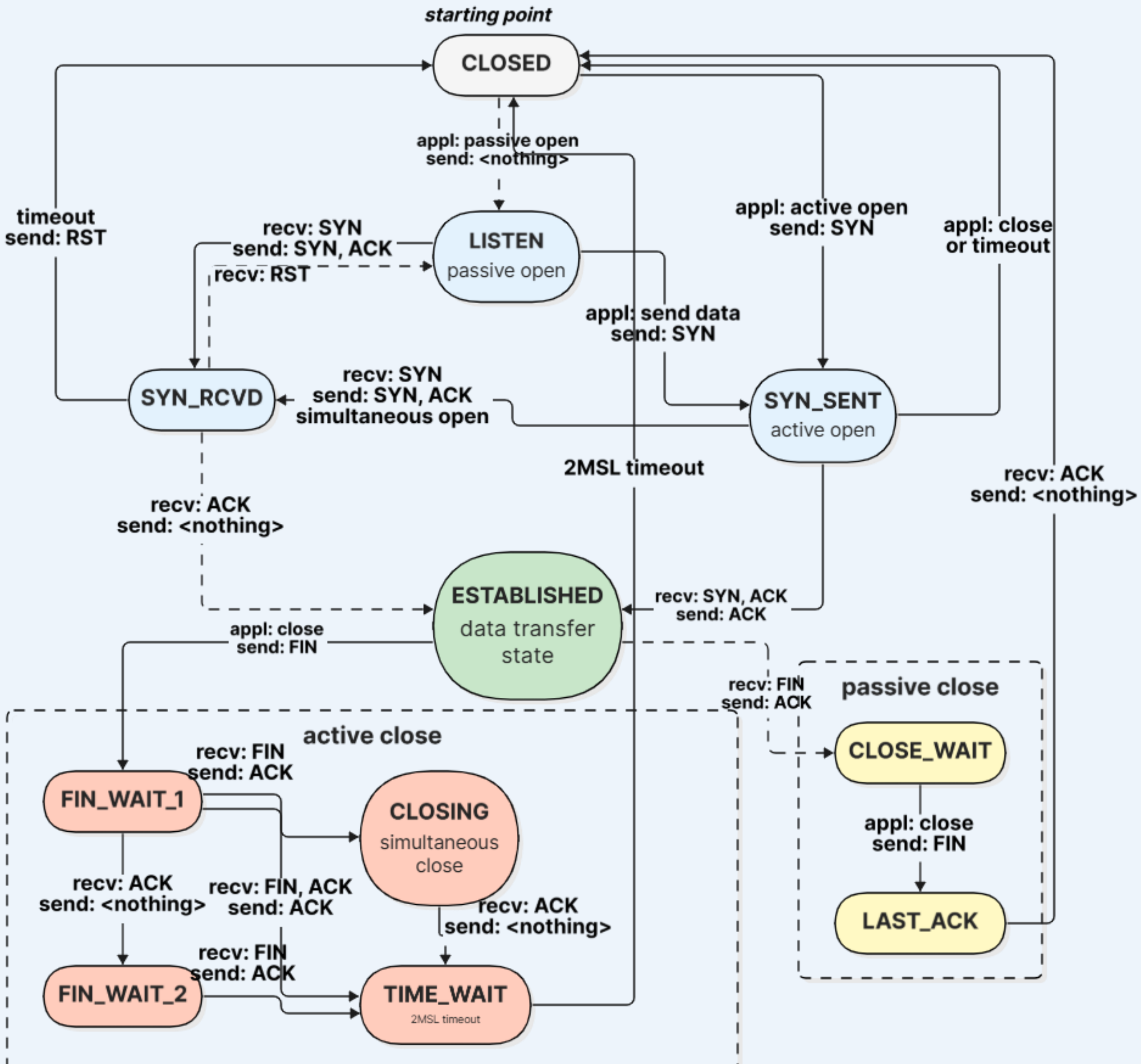
III. **FIN (Finish) from Receiver:** The receiver then sends its own FIN when it is ready to close the connection.

IV. **ACK (Acknowledge):** The sender responds with an ACK, completing the termination.

TCP: STATE TRANSITION

State	Description
CLOSED	There is no connection.
LISTEN	The server is waiting for calls from the client.
SYN-SENT	A connection request is sent; waiting for acknowledgment.
SYN-RCVD	A connection request is received.
ESTABLISHED	Connection is established.
FIN-WAIT-1	The application has requested the closing of the connection.
FIN-WAIT-2	The other side has accepted the closing of the connection.
TIME-WAIT	Waiting for retransmitted segments to die.
CLOSE-WAIT	The server is waiting for the application to close.
LAST-ACK	The server is waiting for the last acknowledgment.

TCP: STATE TRANSITION DIAGRAM



TCP APPLICATION

Web Browsing

HTTP / HTTPS

- Loads web pages completely and in order
- Prevents missing images or broken content

Email

SMTP · IMAP · POP3

- Delivers emails & attachments accurately
- Prevents data loss or corruption

File Transfer

FTP / SFTP

- Reliably transfers large files
- Retransmits lost data for complete delivery

Remote Access

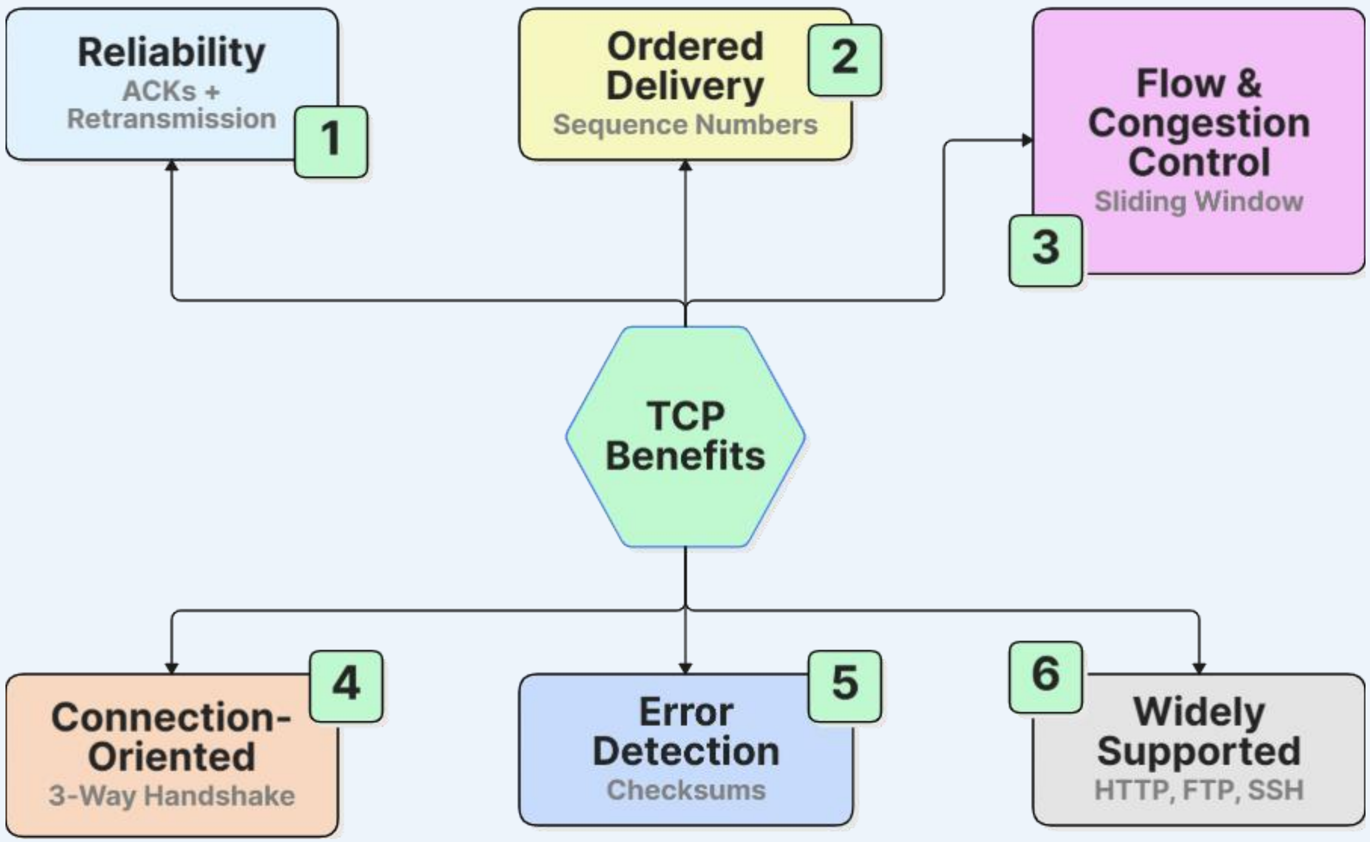
SSH · Telnet

- Reliable command transmission
- Stable, long-lived remote sessions

TCP

Reliable Protocol Powering Everyday Internet Services

TCP ADVANTAGES



TCP LIMITATIONS

Overhead
Handshaking, ACKs & 20+ byte headers add overhead.
1

Slower Speed
Reliability mechanisms add latency vs. UDP.
2

Not Ideal for Real-Time
Retransmission delays hurt streaming, gaming & VoIP.
3

Connection Setup Time
Three-way handshake delays data transfer.
4

Resource Intensive
Buffers, timers & state consume memory and CPU.
5

Head-of-Line Blocking
Lost packet stalls all following packets.
6

REFERENCES

[1] Eddy, W. (2022). Transmission control protocol (TCP) (No. rfc9293).

[2] Davidson, J. (2012). An introduction to TCP/IP. Springer Science & Business Media.

[3] Virginia Tech, *TCP Protocol Lecture Slides (CS4254)*. [Online]. Available: https://courses.cs.vt.edu/cs4254/spring06/slides/TCP.ppt

[4] Tutorials Point. (n.d.). *Transmission Control Protocol (TCP).* Tutorials Point. Retrieved July 15, 2026,

from https://www.tutorialspoint.com/data_communication_computer_network/transmission_control_protocol.htm (TutorialsPoint)

[5] GeeksforGeeks. (2025, October 13). *Transmission Control Protocol (TCP).* GeeksforGeeks. Retrieved July 15, 2026,

from https://www.geeksforgeeks.org/computer-networks/what-is-transmission-control-protocol-tcp/ (GeeksforGeeks)

[6] Eraser. (n.d.). *Eraser: Diagramming for engineering teams*. Retrieved July 16, 2026, from https://app.eraser.io/

[7] VEED. (n.d.). *VEED.IO: Online video editor*. Retrieved July 16, 2026, from https://www.veed.io/

[8] Parziale, L., Liu, W., Matthews, C., Rosselot, N., Davis, C., Forrester, J., & Britt, D. T. (2006). TCP/IP tutorial and technical overview.

[9] R. Tamanna, *TCP/IP Internet Project*, GitHub Repository. [Online]. Available: https://github.com/rubayamist/TCP-IPInterNet

[10] R. Tamanna, *Personal Portfolio*. [Online]. Available: https://rubayamist.github.io/